

NAME: _____

PERIOD: _____

A Controller Comparison

AP Statistics · Unit 4 · Topic 4.8

[STAR] THE PROBLEM

Suppose a food cooperative has 1,200 refrigerated display cases with conventional controllers and 900 with adaptive controllers that adjust cooling automatically. It takes independent random samples of 40 cases from each group; controller type was not assigned. A 95% interval for conventional minus adaptive mean daily use is (1.219, 4.781) kWh per case. Replacing controllers is worthwhile only if mean savings are at least 2.0 kWh.

Rhea: “Zero is excluded and 2 is inside, so replacing controllers is shown to cause savings of at least 2 kWh.”

Sol: “I would compare the interval with both 0 and 2, then check how the groups were formed.”

Daily kWh samples

Controller	N	n	$\bar{x}$	s
Conv.	1,200	40	18.6	4.0
Adaptive	900	40	15.6	4.0

FIRST TAKE — one sentence before you work the parts

Does this range seem to promise savings of at least 2 kWh? Explain in one sentence.

[BOOK] CHECK ZERO, THE CLAIM'S THRESHOLD, AND THE DESIGN (keep on the page)

A confidence interval for $\mu_1 - \mu_2$ describes plausible values for a difference in population means in the stated subtraction order, not a range containing a percent of individual observations. If 0 is excluded, the interval supports a population mean difference; a practical claim such as $\mu_1 - \mu_2 \geq c$ requires the entire interval to lie at or beyond c . For a fixed interval, say “we are confident,” not “there is a probability.” Random sampling supports population scope, while random assignment is needed to isolate a causal treatment effect.

Work (a)–(c).

DOK 1 (a) Locate 0 and 2 relative to the interval.

DOK 2 (b) Interpret the supplied 95% interval in one sentence about the two populations of display cases.

DOK 3 (c) In 3–4 sentences, decide whose account is justified. Use 0 and the 2-kWh cutoff, then explain why the way controllers were chosen limits a cause-and-effect claim.

Frame: Relative to the interval, 0 is _____ and 2 is _____; those positions support _____.

Frame: The sampling design supports _____ but leaves _____ unresolved because _____.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored E / P / I.